

Individual Contribution Statement

Project: Panorama Generation from Overlapping Photos

Course: CO5430: Image Processing (Project ID: P09, Group: G16)

This document outlines the specific contributions made by each group member towards the successful completion of the panorama generation project. The workload was distributed equally, with each member focusing on specific core modules of the pipeline.

1. K.D.N. Chandrasena (E/22/054)

- **Data Collection & Preprocessing:** Captured and organized the custom smartphone image dataset, ensuring appropriate overlap and consistent lighting conditions.
- **Baseline Implementation:** Developed the initial baseline pipeline utilizing the ORB feature detector and Brute-Force matching.
- **Feature Optimization:** Implemented Region of Interest (ROI) based feature extraction and spatial Non-Maximum Suppression (NMS) filtering to improve the spatial distribution of keypoints.

2. K.P.B.P. Karunanayake (E/22/184)

- **Advanced Feature Matching:** Upgraded the pipeline to utilize SIFT features combined with a FLANN-based matcher (KD-Trees).
- **Outlier Rejection:** Integrated Lowe's ratio test to eliminate ambiguous correspondences.
- **Software Architecture:** Led the complete refactoring of the monolithic script into a modular, production-ready architecture, developing the iterative multi-image stitching pipeline.

3. K.J.V. Kahagalla (E/22/179)

- **Global Pose Refinement:** Researched, designed, and implemented the global Bundle Adjustment module using Levenberg-Marquardt optimization to mitigate cumulative alignment drift.
- **Evaluation Framework:** Built the `metrics.py` framework to automatically calculate RMSE and inlier ratios.
- **Documentation:** Led the quantitative analysis and the drafting and formatting of the final 8-page IEEE conference-style report.

4. K.A.H. Kumarasinghe (E/22/205)

- **Geometric Transformations:** Developed the RANSAC homography estimation logic and implemented the innovative "Smart Fallback" mechanism to prevent catastrophic perspective stretching.
- **Image Blending:** Researched and implemented advanced compositing techniques, specifically Distance-Transform Feathering and Laplacian Pyramid Multi-Band Blending.
- **Presentation:** Prepared the final presentation slides and organized the visual demonstrations for the final submission.

All members actively participated in group discussions, code reviews, and the final integration testing of the pipeline.